ASSOCIATE PRODUCT MANAGER (APM) CASE STUDY · FINTECH PRODUCT SYSTEMS

ChargeDossier

Deterministic Decision Support for Payment Dispute Evidence: An Associate Product Manager Case Study in Explainable Fintech Operations

AUTHOR · CANDIDATE
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TARGET ROLE · COMPANIES
General APM · Mastercard, Visa, Razorpay, Fintechs

VERIFIED TEST SUITE
`24/24 Tests Passed (100%)`

PUBLISHED ARTIFACTS
Live Demo · Case Study · Source

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Engineering Discipline Meets Epistemic Honesty

What problem was explored, what prototype was engineered, what was verified, and what remains an unvalidated hypothesis.

1. THE PROBLEM

High-Ticket Margin Drag

Merchants selling $150–$2,000+ goods face polarized options: 30–60 min manual assembly across 4 disconnected SaaS tools (sub-20% win rate), or autopilot bots charging a **25% contingency take-rate** that submit fatal contradictions blindly.

$75–$375

FEE DRAG PER $300–$1.5K CASE

2. WHAT WAS BUILT

Deterministic Review Copilot

A zero-dependency evidence auditor scoring dispute readiness (0–100) across 5 rule categories modeled on Mastercard 4837/4853 heuristics, flagging fatal flaws, compiling Mastercom-compliant dossiers, with an explicit human approval lock.

< 90 Sec

EVIDENCE TRIAGE WINDOW

3. WHAT WAS VERIFIED

24 Automated Tests Passed

Verified across 4 distinct synthetic high-ticket edge cases in native Node.js (312ms runtime). Dual execution verified: local REST API and 100% in-browser client execution on live GitHub Pages with zero server dependencies.

100%

TEST PASS RATE (3 SUITES)

4. UNVALIDATED CLAIMS

Epistemic Boundaries

Customer willingness to pay $49/mo flat SaaS vs. 25% contingency remains unvalidated. Operator review friction (60–90 sec) vs. autopilot preference is unproven. **Zero customer interviews conducted** (outreach paused awaiting review).

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UNVERIFIED COMMERCIAL CLAIMS

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The Three Evidence Failure Modes in High-Ticket Disputes

Why online merchants write off winnable disputes and why blind submission causes permanent losses.

FAILURE MODE A — MISSING

Absent Critical Exhibits

Under scheme guidelines, disputes require specific nexus proofs. On Reason Code 4853 (Merchandise) claims, submitting tracking without attaching the store's **published return policy** or customer support remedy results in automatic issuing bank rejection.

- Missing timestamped checkout terms acceptance
- Absent carrier signature receipt on orders > $150
- No documented repair or replacement offer transcript

FAILURE MODE B — CONTRADICTORY

Self-Defeating Evidence

Automated bots ingest data without semantic cross-checks. Submitting carrier tracking showing a package is still **"In Transit"** (weather delay) or an order delivered to a Florida freight forwarder with failed AVS guarantees rejection while wasting administrative fees.

- Package delivered to unverified forwarder destination
- Support transcript contains cardholder theft claim
- Billing state mismatch with failed AVS street match ('N')

FAILURE MODE C — IRRELEVANT

Scheme Mismatch

Evidence that defeats one reason code is useless for another. Submitting terms of service to counter Reason Code 4837 (Fraud / No Authorization) fails because cardholders did not authorize checkout. Only 3DS tokens, AVS/CVC matches, and device IP establish liability shift.

- Survey: ~60% of merchants abandon 40%+ chargebacks [S-15]
- Manual fraud win rate averages under 20% [S-06]
- LexisNexis: $1 fraud costs $4.61–$5.13 in total friction [S-05]

Sources: LexisNexis [S-05] · MRC / Cybersource [S-06] · Riskified / Datos Insights [S-15]

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The Product Hypothesis: Pre-Flight Triage & Human Control

Replacing 25% contingency bots and opaque LLMs with an explainable, deterministic copilot.

THE CORE HYPOTHESIS

"Direct-to-consumer merchants selling high-ticket physical goods ($150–$2,000+ AOV) will adopt an upfront, explainable review tool with human decision control at $49/mo in preference to a 25% contingency take-rate, provided evidence triage takes < 90 seconds."

Economic Comparison on $1,000 Dispute:

- *Contingency Bot:* Merchant pays $250 fee on recovery ($750 net). Blind submission risks submitting fatal flaws.
- *ChargeDossier Copilot:* Flat SaaS ($49/mo). Pre-flight audit surfaces missing return policy, elevates evidence score from 73 to 100, and recovers $1,000 ($951 net on 1 dispute).

$201 Net Gain

THEORETICAL ADVANTAGE ON SINGLE $1,000 WIN

ARCHITECTURAL PRINCIPLES

Why Deterministic Math Beats Generative AI

In financial dispute representment, submitting fabricated or hallucinated statements to issuing banks risks scheme fines and acquiring account termination. ChargeDossier rejects black-box LLMs in favor of strict deterministic math.

- **100% Mathematical Auditability:** Every rule assigns verifiable points and reasons based on published scheme standards.
- **Zero Hallucination Risk:** The engine never invents customer quotes, fake tracking numbers, or unrecorded policies.
- **Early Loss Acceptance as a Win:** Rejecting unwinnable claims saves operator time and avoids wasting staff hours on unviable submissions.
- **Explicit Status:** This thesis is an exploratory product hypothesis, not a confirmed commercial outcome.

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The 7-Step Evidence Auditing & Dossier Lifecycle

From disconnected merchant silos to a Mastercom-modeled Second Presentment submission in under 90 seconds.

STEP 01

Ingest & Inspect

4-tab inspection pane displays Order details, Gateway Auth (AVS/CVC/3DS), Carrier Tracking (POD), and Policy transcripts.

STEP 02

Deterministic Audit

One-click evaluation assesses evidence against 5 heuristic rule categories modeled on Mastercard 4837/4853 standards.

STEP 03

Flag Contradictions

Automated checks catch fatal flaws: package in-transit, destination mismatch, or freight forwarder fraud patterns.

STEP 04

Interactive Fix

Actionable checklist allows operator to attach missing warranty policy and support transcripts, elevating score dynamically.

STEP 05

Compile Dossier

Compiles Mastercom-compliant executive rebuttal letter and structured Exhibits A–E (<18 pages, <10MB layout constraints).

STEP 06

Human Review Lock

Operator checkpoint to explicitly **Approve** viable claims or **Reject** unviable claims with logged notes.

STEP 07

Instant Demo Reset

One-click reset clears all decision locks and restores all 4 synthetic records back to baseline unreviewed state.

Verified Workflow: All 7 steps execute locally and on live GitHub Pages sandbox

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Zero-Dependency Engineering & Dual Execution

Modular ES6 architecture running both as a local Node.js server and 100% in-browser on GitHub Pages.

IMPLEMENTED COMPONENTS

PROTOTYPE CODEBASE

- **Deterministic Rules Engine (`src/engine.js`):** Pure functional rule evaluation, mathematical deduction scoring, and tier classification. Zero external dependencies.
- **State Machine (`src/state.js`):** Governs lifecycle mutations: `UNREVIEWED → AUDITED → (APPROVED | REJECTED)` with full reset capabilities.
- **Dossier Generator (`src/generator.js`):** Formats Mastercom-modeled Second Presentment rebuttal letters and structured Exhibits A through E.
- **Native HTTP Server (`server.js`):** Zero-dependency Node.js HTTP server routing REST endpoints and serving static assets.
- **Standalone Client Controller (`app.js`):** Drives the dashboard in-browser via pure ES modules, executing seamlessly on GitHub Pages.

FUTURE ENTERPRISE INTEGRATIONS

ROADMAP

- **Mastercard Mastercom API v6 Connector:** Direct licensed acquirer submission endpoint complying with Mastercom Claim ID tracking and 10MB/18-page limits [S-01].
- **Payment Gateway Connectors:** Direct programmatic evidence upload via Stripe REST API (`POST /v1/disputes/:id`) and Shopify Payments webhooks.
- **Carrier POD Webhooks:** Automated real-time delivery GPS and signature ingestion from FedEx Ship Manager and UPS Tracking APIs.
- **Zendesk / Gorgias Support Ingestion:** OAuth ticket transcript extraction and customer dispute deflection logging.

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EXTERNAL NPM PACKAGES INSTALLED

Engineering Discipline: Pure Node.js & Vanilla ES6 · 24 Passing Automated Tests

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The Deterministic Decision Framework

Actual point allocations, contradiction penalties, and quality tier boundaries from `src/engine.js`.

GROSS SCORING (100 PTS)

Heuristic Rule Allocations

- **Payment & Identity (30 pts):** AVS street/zip match (12 pts), CVC match (8 pts), 3DS authentication (10 pts).
- **Carrier POD & Tracking (35 pts):** Delivered status (15 pts), destination match (10 pts), cardholder signature on ≥ $150 (10 pts).
- **Policies & Disclosures (15 pts):** Terms accepted at checkout (8 pts), published return policy attached (7 pts).
- **First-Party Trust (20 pts):** Prior orders & account tenure (10 pts), customer communication/remedy logs (10 pts).

CONTRADICTION DEDUCTIONS

Fatal Evidentiary Flaws

- **Delivery Destination Mismatch:** Package delivered to different address → **-20 pts** (Critical)
- **Third-Party Signature:** Signed by freight forwarder, not customer → **-10 pts** (Warning)
- **Explicit Fraud Affirmation:** Cardholder stated card was stolen → **-25 pts** (Critical)
- **Defect Without Remedy:** Merchandise defect without repair offer → **-10 pts** (Warning)
- **State Mismatch + Failed AVS:** Billing state ≠ Shipping with AVS 'N' → **-20 pts** (Critical)

DECISION TIERS

Explainable Recommendations

- **TIER A (80–100 Pts):** Strong evidence, zero contradictions, zero missing items → `APPROVE_FOR_PRESENTMENT`
- **TIER B (60–79 Pts):** Fixable gaps detected, no critical contradictions → `REVIEW_AND_ATTACH_EVIDENCE`
- **TIER C (0–59 Pts):** Deficient evidence or critical contradictions → `REJECT_AND_ACCEPT_LOSS`

`Net = Gross - Penalties`

STRICT MATHEMATICAL FORMULA

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Synthetic Benchmark Scenarios & Exact Score Outputs

Tested against 4 realistic high-ticket payment dispute edge cases modeling Mastercard reason codes.

CASE ID	PRODUCT & AMOUNT	REASON CODE	EVIDENTIARY CONDITION	INITIAL SCORE	REMEDIATION	FINAL DECISION
DSP-1049	**AeroSound Pro ANC Headphones** $349.00 USD	MC 4837 (FRAUD)	Full AVS (Y), CVC (M), 3DS Auth, FedEx signature slip (E. Vance), Zendesk usage confirmed.	100 / 100 TIER A	None Needed	**Approved** Locked for submission
DSP-2081	**BaristaCraft Burr Grinder** $295.00 USD	MC 4853 (NOT RECEIVED)	Carrier tracking confirms *"In Transit — Weather Delay in Spokane"*. Unsigned package.	22 / 100 TIER C	Unviable	**Rejected (Loss Accepted)** Saves operator labor
DSP-3312	**ErgoLift Pro Standing Desk** $850.00 USD	MC 4837 (FRAUD)	TX billing vs. FL shipping. Failed AVS (N), 3DS failed, forwarder signature, customer fraud call.	2 / 100 TIER C	Unviable	**Rejected (Loss Accepted)** 3 critical contradictions caught
DSP-4190	**Modular Platform Bed Frame** $1,150.00 USD	MC 4853 (DEFECTIVE)	Delivered with signature. Return policy missing and defect reported without remedy log.	73 / 100 TIER B	**Attached Policy & Remedy** +27 Pts Recalculated	**Elevated to 100 (Tier A)** Approved after fix

Exact values verified from src/data/disputes.js · Zero synthetic PII used.

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24 Passing Automated Tests & In-Memory Test Harness

Reproducible test suite verifying unit rules, REST API routes, and full state machine workflows in 312ms.

SUITE 1: RULES ENGINE — 7 TESTS PASSING

Scoring & Resilience

- DSP-1049 qualifies as TIER A with score ≥ 85
- DSP-2081 qualifies as TIER C (missing POD)
- DSP-3312 triggers severe contradiction penalties
- DSP-4190 qualifies as TIER B with actionable fixes
- Defensive null/undefined input validation
- Score boundary tests (80 Tier A, 79 Tier B, 59 Tier C)
- Defensive resilience on missing payload objects

SUITE 2: REST API SERVER — 13 TESTS PASSING

HTTP In-Memory Router

- `GET /api/disputes` returns 4 dispute records
- `GET /api/disputes/DSP-1049` returns details
- `POST /audit` evaluates evidence to TIER A
- `POST /dossier` generates Second Presentment
- `POST /approve` locks state to APPROVED
- `POST /reject` locks state to REJECTED
- `POST /fix` applies remediation and updates score
- `POST /reset` restores initial state
- OPTIONS CORS preflight handling & 404 handling

SUITE 3: E2E WORKFLOWS — 4 TESTS PASSING

Lifecycle Verification

- **Workflow 1 (Happy Path):** Ingest → Audit → Generate Dossier → Approve (DSP-1049)
- **Workflow 2 (Deficient Rejection):** Audit → Catch Contradiction → Reject (DSP-2081)
- **Workflow 3 (Gap Remediation):** Ingest → Audit (73) → Fix Policy → Score Elevates (100) → Approve (DSP-4190)
- **Workflow 4 (Demo Reset):** Mutate records → Reset → All 4 restored to UNREVIEWED

Tests: node --test tests/*.test.js (24/24 pass) · Benchmark: node tests/benchmark.js (~0.01ms latency, $\sigma^2 = 0.0000$)

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Primary Regulatory & Industry Source Mapping

Catalog of 25 primary and secondary industry references supporting system design.

CARD SCHEME RULES

Mastercom & FPT

Mastercom API v6 [S-01]: Enforces strict technical limits (<10MB file, <18 pages).

First-Party Trust [S-02]: Establishes Reason Code 5002 (First-Party Trust Reject) to deflect invalid 4837 fraud claims via account tenure and delivery logs.

REGULATORY ANALYSIS

CFPB Section 1033

Statutory Text [S-16]: CFPB Section 1033 explicitly excludes commercial and business bank accounts from open banking mandates.

Strategic Call: Redirected discovery to payment dispute data where merchants hold direct legal standing.

INDUSTRY BENCHMARKS

Empirical Baselines

MRC & Cybersource [S-06]: Manual fraud chargeback win rates average below 20%.

Riskified & Datos [S-15]: ~60% of merchants abandon 40%+ disputes.

LexisNexis [S-05]: $1 fraud costs $4.61–$5.13 in total merchant friction.

COMMERCIAL TENSIONS

Alternative Pricing

Chargeflow [S-11]: Advertised 25% contingency fee on gross recovery creates severe margin drag on high-ticket goods ($250 on $1k).

Chargebacks911 [S-13]: $1,500+/mo retainers prohibit small/mid-market D2C merchants.

Documented in full in research/source-register.md (25 cataloged sources)

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Limitations & The Phase 2B Validation Roadmap

Explicit boundary delineation and empirical next steps required before commercialization.

HONEST PROTOTYPE LIMITATIONS

NOT VALIDATED

- **Customer Willingness to Pay ($49/mo) is Unproven:** While merchants object to 25% contingency rates, their behavioral preference might still favor "no-win, no-fee" models over recurring SaaS.
- **Operator Tolerance for Review Friction:** Whether busy merchant operators are willing to spend 60–90 seconds auditing cases vs. demanding 100% hands-off automation is an unvalidated hypothesis.
- **Zero Live Merchant Interviews Conducted:** Outreach was paused awaiting explicit authorization. No live customer interviews have occurred.
- **Synthetic Sandbox Only:** Prototype operates on synthetic benchmarks; no live payment gateway or card network connections exist.

3-PHASE VALIDATION ROADMAP

60-DAY EXECUTION

- **Phase 1 — Discovery Interviews ($n=8$):** Execute semi-structured interviews with verified operations/risk leads at US high-ticket D2C merchants ($150–$1,500 AOV) using documented prospecting kit.
- **Phase 2 — Blinded Shadow Evaluation:** Ingest 10 redacted historical disputes from 3 pilot merchants. Compare ChargeDossier shadow drafts against original manual/bot submissions.
- **Phase 3 — Open Benchmark Dataset:** Release open, anonymized dispute evidence dataset to advance academic and industrial research on explainable dispute representment.

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ChargeDossier: Verified Artifacts & Role Alignment

Summary of product contributions and live public verification links.

GENERAL APM COMPETENCY ALIGNMENT

MASTERCARD · VISA · RAZORPAY · FINTECHS

- **Problem Discovery & Customer Sizing:** Quantified merchant friction via empirical industry data ($4.61–$5.13 fraud cost multiplier [S-05], 60% abandonment [S-15]) and identified high-ticket AOV contingency margin drag.
- **Product Hypothesis & User Empathy:** Defined narrow ICP ($150–$1,500 AOV D2C brands), formulated the flat $49 SaaS vs. 25% contingency trade-off, and established clear falsification criteria.
- **Prioritization & Technical Trade-offs:** Chose deterministic heuristic rules over generative LLMs to eliminate hallucination liability in adversarial compliance; built zero-dependency architecture executing in <1ms.
- **Workflow Design & Metrics:** Designed human-in-the-loop triage balances automated exhibit assembly with strategic operator approval; established Net Merchant Recovery as primary north star.
- **Product Strategy Across Payment Networks:** Operationalized scheme rules (Mastercard Mastercom/FPT, Visa CE3.0, gateway dispute fees) balancing issuer, acquirer, and merchant incentives.

VERIFIED PUBLIC DELIVERABLES

ALL LIVE

- **Live Interactive Demo** (In-Browser Sandbox) →
- **Dedicated Portfolio Case Study** (Aura Portfolio) →
- **Portfolio Projects Showcase** (Aura Portfolio) →
- **GitHub Source Code** (24 Passing Tests) →

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